

LoRa Communication System for Exploratory Rovers: Viability Assessment and Performance Analysis

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Abstract—Mars rovers require reliable, low power communication for telemetry and command transmission. This paper evaluates the viability of LoRa for rover to base station communication through range, spreading factor, and power consumption experiments. Testing measured RSSI, SNR, packet delivery rate, and energy usage at distances up to 1000 ± 5 m. Results show an average 99% packet delivery at 1000 m (-122.94 ± 2.31 dBm RSSI, 2.51 ± 2.81 dB SNR). SF12 achieved 1000 ± 5 m range at 91.8 ± 1.3 mA transmit current. Average operating current ranged from 32.8 ± 1.2 mA at SF7 to 80.6 ± 1.2 mA at SF11; SF7 enabling 39.6 ± 2.4 hours on a $1300 \pm 5\%$ mAh battery, whereas SF11 achieving 16.1 ± 0.8 hours. These results indicate that LoRa is suitable for multi-kilometre, energy-constrained (1300 mAh battery) rover missions.

I. INTRODUCTION

Planetary surface missions impose strict constraints on communication system design, primarily due to limited onboard power, absence of supporting infrastructure, and propagation losses over long-range wireless links. Robotic platforms such as Mars rovers therefore require lightweight, energy efficient communication systems capable of sustaining reliable telemetry and command exchange under constrained energy budgets.

Conventional short-range wireless protocols such as Wi-Fi and Bluetooth are optimised for local connectivity and are generally not suitable for rover communications. Their limited range and, in the case of Wi-Fi, comparatively higher transmission power requirements [1] restrict their applicability in low infrastructure, long-distance robotic deployments.

Low Power Wide Area Network (LPWAN) technologies, particularly LoRa (Long Range), provide an alternative approach for long-range, low-power communication. LoRa employs chirp spread spectrum (CSS) modulation to achieve high receiver sensitivity and extended communication range at low data rates. Prior studies have demonstrated reliable LoRa communication over several kilometres, with performance strongly dependent on spreading factor [2] [3] [4]. These characteristics make LoRa a candidate technology for energy-constrained robotic communication.

This paper evaluates LoRa-based rover-to-base communication using an Arduino MKR WAN 1310 platform. Experimental measurements assess range, packet delivery ratio

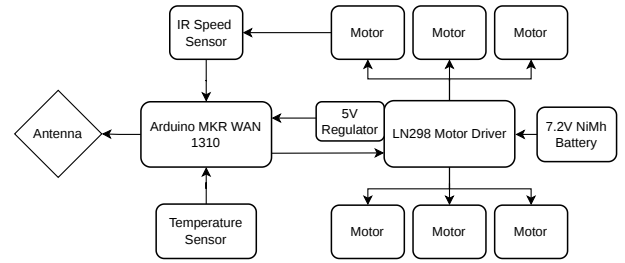


Fig. 1. High-level architecture of the Mars rover showing main subsystems and their interconnections.

(PDR), received signal strength indicator (RSSI), signal-to-noise ratio (SNR), and energy consumption across multiple spreading factor configurations. The results quantify the trade-offs between reliability, range, and energy efficiency in a rover relevant deployment scenario.

The remainder of this paper is structured as follows: Section II presents the system architecture; Section III describes the experimental methods, results and analysis; Section IV discusses the findings; and Section V concludes the study.

II. SYSTEM OVERVIEW

A. Rover System Architecture

The rover system comprises four primary subsystems, as illustrated in Fig. 1: power, sensing, control, and communication. The MKR WAN 1310 coordinates all subsystems, processing sensor data and issuing motor commands while managing the LoRa communication link. Power is supplied by a 7.2 V, 1300 mAh NiMH battery, regulated to supply each subsystem within its operating voltage range.

B. LoRa Communication Subsystem

The communication subsystem is implemented using the Arduino MKR WAN 1310, operating at 868 MHz in the EU ISM band. The module employs LoRa modulation with configurable spreading factors (SF7–SF12), 125 kHz bandwidth, and a coding rate of 4/5. Bidirectional communication is supported,

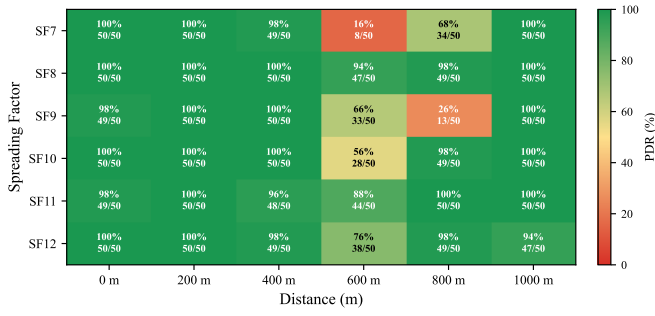


Fig. 2. Packet delivery heat map across SF7–SF12 over 0–1000 m range.

with the rover transmitting telemetry packets to a fixed base station and receiving command messages in return.

C. Communication Requirements

The communication subsystem must achieve a minimum reliable range of 1000 m with a PDR exceeding 95%, and support a transmission rate of 1 Hz to provide adequate telemetry updates for rover operation. These requirements drive the experimental evaluation presented in Section III.

III. EXPERIMENTAL METHODS, RESULTS AND ANALYSIS

A. Range Performance Testing

The range test was conducted to assess LoRa link reliability as a function of distance and spreading factor. Measurements were performed in an open field environment with minimal vegetation and nearby buildings to reduce multipath reflections and obstruction losses [5]. This site also approximated the low obstruction terrain conditions relevant to extraterrestrial surfaces. Testing was carried out to a maximum distance of 1000 ± 5 m in 200 ± 5 m increments, with transmitter separation verified using GPS measurements.

The experimental setup used two identical Arduino MKR WAN 1310 units connected to laptop computers for data logging and control. Each node was fitted with a 0 dBi gain dipole antenna [6]. The transmitter sent 50 packets at each test location, cycling from SF7 to SF12 at a transmission rate of 1 Hz with a 4 byte payload. Each packet contained a packet identification number; the receiver logged the corresponding RSSI and SNR values upon successful reception. All tests used 868 MHz, 125 kHz bandwidth, coding rate 4/5, and 14 dBm transmit power.

1) *Range Results:* Range performance did not degrade uniformly with distance. Instead, the results indicate threshold dominated behaviour: PDR remained close to 100% (Fig. 2) up to approximately 600 m, followed by a sharp reduction in reliability despite relatively stable RSSI values (Fig. 3).

This suggests that received signal strength alone is not the primary limiting factor. RSSI remained broadly consistent with the Free Space Path Loss (FSPL) model across the tested range, but with an approximate 60 dB offset from the FSPL model, indicating that path loss alone cannot explain the observed link failures within the measurement environment.

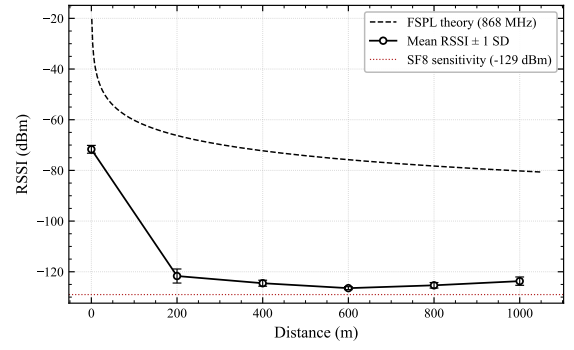


Fig. 3. RSSI variation with distance at SF8.

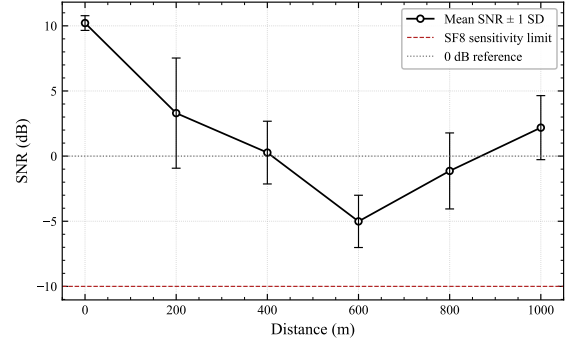


Fig. 4. SNR variation with distance at SF8.

The free space path loss (FSPL) model was calculated by

$$\text{FSPL}(d, f) = \left(\frac{4\pi df}{c} \right)^2 \quad (1)$$

where d is the separation distance (m), f is the carrier frequency (Hz), and c is the speed of light (3×10^8 m s⁻¹). Expressed in decibels, (1) becomes

$$\text{FSPL}_{\text{dB}} = 20 \log_{10}(d) + 20 \log_{10}(f) - 147.55 \quad (2)$$

where the constant $-147.55 = 20 \log_{10}(4\pi/c)$.

The received signal power P_r is then estimated using the link budget

$$P_r = P_t + G_t + G_r - L_c - \text{FSPL}_{\text{dB}} \quad (3)$$

where P_t is the transmit power (dBm), G_t and G_r are the transmit and receive antenna gains (dBi) respectively, and L_c accounts for cable and connector losses (dB).

In contrast, SNR declined at 600 m and 800 m, reaching values of -5 dB and -1.1 dB respectively for SF8, approaching the demodulation threshold (approximately -10 dB) required for successful decoding. This indicates that packet loss was more strongly associated with noise and interference than with average received power. The mean SNR for SF7 (PDR of 16%) at 600 m was -7.84 dB, indicating further that the packet loss is correlated to SNR.

TABLE I
LoRA TRANSMISSION CURRENT AND TIME-ON-AIR BY SPREADING FACTOR

SF	Meas. (ms)	Theo. (ms)	Δ (%)	I_{TX} (mA)	I_{idle} (mA)
SF7	70	72	2.7	91.8±1.7	28.4±1.3
SF8	120	134	10.2	91.2±1.5	29.3±1.1
SF9	230	247	7.0	91.4±1.6	29.4±1.1
SF10	451	453	0.3	91.6±1.5	29.2±1.1
SF11	825	823	0.2	91.5±1.5	29.2±1.0
SF12	1647	1647	0.1	91.8±1.3	29.1±1.1

TABLE II
ESTIMATED BATTERY LIFE PER SPREADING FACTOR (1300±65 MAH, 1 HZ CYCLE)

SF	δ	I_{avg} (mA)	t_{batt} (h)
SF7	0.070	32.8±1.2	39.6±2.4
SF8	0.120	36.7±1.0	35.4±2.0
SF9	0.230	43.7±0.9	29.8±1.6
SF10	0.451	57.4±0.9	22.7±1.2
SF11	0.825	80.6±1.2	16.1±0.8

Notably, PDR recovered to approximately 99% at 1000 m. As shown in Fig. 2, the corresponding SNR rises above the local noise floor, suggesting that the earlier reduction was caused by site specific interference rather than absolute range limitation. The maximum viable communication distance is therefore likely much greater than 1 km under favourable conditions.

B. Power Consumption

Power consumption testing was conducted to evaluate the transmission energy requirements of the LoRa system under different spreading factor settings. The objective was to quantify the effect of spreading factor and payload size on battery life.

An Arduino MKR WAN 1310 was supplied from a regulated 5 V source. Current was measured using a 1 Ω precision shunt resistor in series with the supply line; the voltage drop across the shunt was monitored with an oscilloscope, allowing instantaneous transmission current to be determined via Ohm's law. The transmitter sent 32-byte packets at SF7–SF12. For each spreading factor, the oscilloscope captured the transmission pulse waveform, from which packet transmission duration and peak/average current were obtained. All tests used 868 MHz, 125 kHz bandwidth, coding rate 4/5, and 14 dBm transmit power.

1) *Power Results:* Table I summarises the measured transmission current, idle current, and packet time on air for each spreading factor. The measured airtime closely follows theoretical LoRa predictions, with errors below 10.2% across all cases and below 1% for SF10–SF12, confirming that the experimental setup captured radio timing behaviour with acceptable accuracy.

Transmit current remained effectively constant at approximately 91–92 mA for all spreading factors, consistent with the RF power amplifier current draw being independent of spreading factor under fixed output power conditions. However, packet airtime nearly doubled with each successive

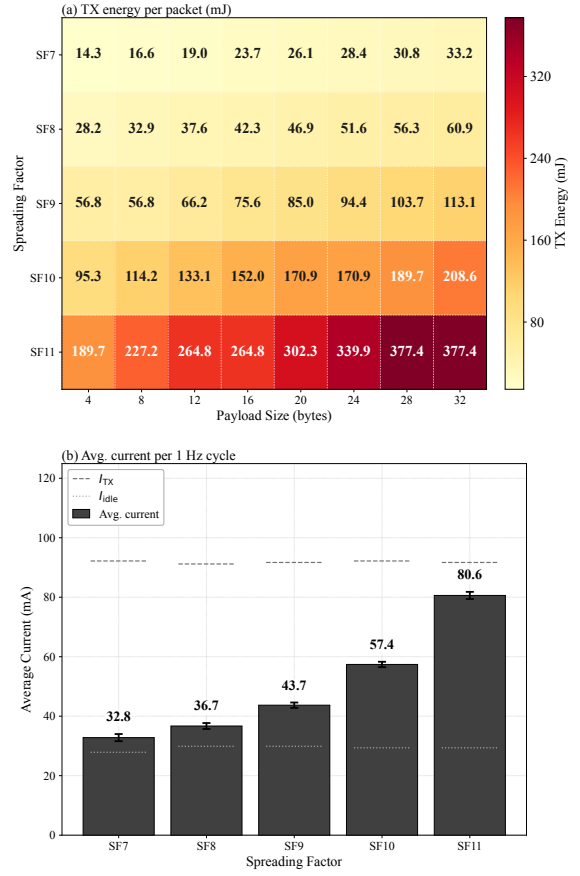


Fig. 5. Transmission energy analysis: (a) energy per packet (mJ) as a function of spreading factor and payload size; (b) average current draw at 1 Hz transmission rate across SF7–SF11.

spreading factor, rising from 70 ms at SF7 to 1647 ms at SF12. Consequently, although instantaneous transmit current is similar across all SF settings, higher spreading factors incur substantially greater total energy consumption.

Fig. 5(a) presents transmission energy per packet for payload sizes between 4 and 32 bytes. Energy consumption rises with both payload size and spreading factor. At SF7, a 32-byte transmission required 33.2 mJ, whereas the same payload at SF11 required 377.4 mJ; over eleven times greater. This confirms that spreading factor selection and payload optimisation are the principal determinants of communication energy cost.

Packet energy E was calculated as

$$E = V \cdot I_{avg} \cdot T_{air}, \quad (4)$$

where V is the supply voltage, I_{avg} is the measured average transmission current, and T_{air} is the packet airtime [7]:

$$T_{air} = (N_{pre} + 4.25 + N_{sym})T_{sym}, \quad T_{sym} = \frac{2^{SF}}{BW} \quad (5)$$

$$N_{sym} = 8 + \max\left(\left\lceil \frac{8PL - 4SF + 28}{4(SF - 2DE)} \right\rceil (CR + 4), 0\right) \quad (6)$$

where BW is bandwidth, PL is payload length in bytes, CR is the coding rate (1 for 4/5), and DE is the low-data-

rate optimisation flag (1 for SF11–SF12, else 0). N_{pre} is the programmed preamble length, CRC and explicit header are assumed throughout. Computed airtimes were verified against oscilloscope measurements before applying (4) to generate Fig. 5(a).

Fig. 5(b) shows the calculated average current for periodic 1 Hz transmissions. The duty cycle δ , defined as the fraction of each packet time on air t_{ToA} over the transmission cycle T_{cycle}

$$\delta = \frac{t_{ToA}}{T_{cycle}}. \quad (7)$$

The average current I_{avg} over one cycle is the duty-cycle-weighted sum of transmit I_{TX} and idle current I_{idle} :

$$I_{avg} = I_{TX} \delta + I_{idle} (1 - \delta). \quad (8)$$

For a battery capacity, C of 1300 ± 65 mAh, estimated battery life is

$$t_{batt} \text{ (hours)} = \frac{C}{I_{avg}}. \quad (9)$$

Average current increased from 32.82 ± 1.22 mA at SF7 to 80.61 ± 1.29 mA at SF11, reducing estimated battery life from 39.61 ± 2.47 h to 16.13 ± 0.85 h in Table II (see uncertainty propagation in Section *Uncertainty Analysis*). The dominant parameter is transmit duty cycle, which increases directly with packet airtime.

C. Uncertainty Analysis

All uncertainties were calculated using Python (Pandas/NumPy) or Excel and are reported as one standard deviation (1σ) for each spreading factor RSSI and SNR. Transmit and idle currents were extracted from oscilloscope CSV files (`sf7.csv`–`sf12.csv`):

Average current uncertainty was propagated as:

$$\sigma_{I_{avg}} = \sqrt{(\delta \sigma_{I_{TX}})^2 + ((1 - \delta) \sigma_{I_{idle}})^2 + ((I_{TX} - I_{idle}) \sigma_{\delta})^2}$$

where $\sigma_{\delta} = 0.00005$ (0.05 ms timing uncertainty over 1 s cycle). Battery life assumed 1300 ± 65 mAh (5% tolerance):

$$\frac{\sigma_t}{t} = \sqrt{\left(\frac{\sigma_C}{C}\right)^2 + \left(\frac{\sigma_{I_{avg}}}{I_{avg}}\right)^2}$$

Results are summarised in Table II.

IV. DISCUSSION

The experimental results demonstrate that LoRa meets the communication requirements defined in Section II-C across spreading factor 7-11 configurations. SF12 does not meet the 1 Hz transmission rate due to time on air reaching 1647ms for a 32-byte packet. A PDR exceeding 99% at 1000 m confirms that the 95% reliability threshold is achievable at the target range. The unexpected dip in PDR at 600–800 m, attributed to site specific interference rather than path loss, highlights the importance of environment characterisation when planning rover deployments. On extraterrestrial surfaces with lower

atmospheric noise floors, link performance is expected to be more uniform with distance.

The trade-off between spreading factor, battery life, and range has clear engineering implications. SF7–SF9 offer significantly extended battery life (29.5–40.1 h) and are preferable where link margin is sufficient. SF10–SF12 provide enhanced receiver sensitivity and link robustness [8] at the cost of substantially increased energy consumption, and should be reserved for obstructed or long range scenarios where reliability is prioritised.

A limitation of this study is the terrestrial test environment, which differs from the conditions in atmospheric composition, pressure, and noise characteristics. The 868 MHz EU ISM band frequency was used for convenience; actual rover deployments may require a different frequency allocation.

V. CONCLUSION

This paper has evaluated LoRa as a communication technology for energy- constrained rover missions. Range testing confirmed a PDR exceeding 99% at 1000 m, satisfying the 95% reliability requirement. SNR analysis identified noise and interference, rather than path loss as the dominant cause of link degradation, with local interference likely responsible for the observed PDR reduction at intermediate distances. Power consumption measurements showed that transmit current remains constant across spreading factors at approximately 92 mA, while airtime and duty cycle govern battery life, yielding estimates of 16.1 ± 0.8 h at SF11 to 39.6 ± 2.4 h at SF7 on a 1300 mAh cell. These results establish LoRa as a viable communication subsystem for kilometre scale short range links, time constrained (20-30 hours) planetary rover applications, with spreading factor selection as the key design parameter balancing range and endurance.

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